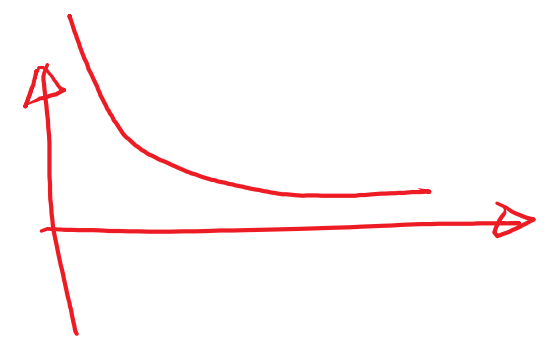
S مسائل	Stable پایداری	عملکرد	ورودی X	{0.001 ,0.003} {0.01 ,0.03} Learning_rate	گیر کردن در Local Min	(تدریجی) (Solver)
رگرسیون (منسوخ)	 ✓	سرعت کند دقت ×	all حجم داده	ثابت	×	گرادیان کاهشی Gradient Descent GD

(مبتنی بر گرادیان)

رگرسیون خطی

$y = a \cdot X + b$

dot ضرب

وزن → 10 ریاضی, 5 علوم, 3 زبان

$$\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix} \cdot \begin{bmatrix}10\\ 5\\ 3\end{bmatrix} = \underbrace{\begin{bmatrix}(1\times10)+(2\times5)+(3\times3)\\ (4\times10)+(5\times5)+(6\times3)\end{bmatrix}}_{(2\times1)} + \begin{bmatrix}2\\ 2\end{bmatrix}$$

مجموع نمرات

(2×3) (3×1) → $[\,\square\,]$

تعداد ستون == تعداد سطر

تعداد سطر

تعداد ستون

Learning rate

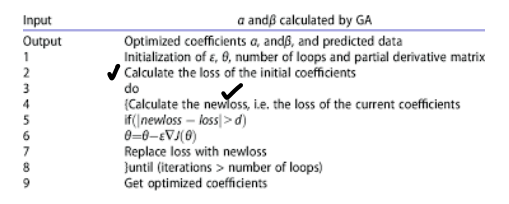
$a = 0$ $b = 0$

$\text{parameters} = \begin{bmatrix} a = 0 \\ b = 0 \end{bmatrix}$

```
for i in range(max_iter):
    y' = a . X + b
    error = y' - y
    ~~Gradient = X.T @ error / n~~
    ~~parameters = parameters - Gradient * Learning_rate~~
    mse(y, y') ──> Cost_function
```

Parameters [0] a

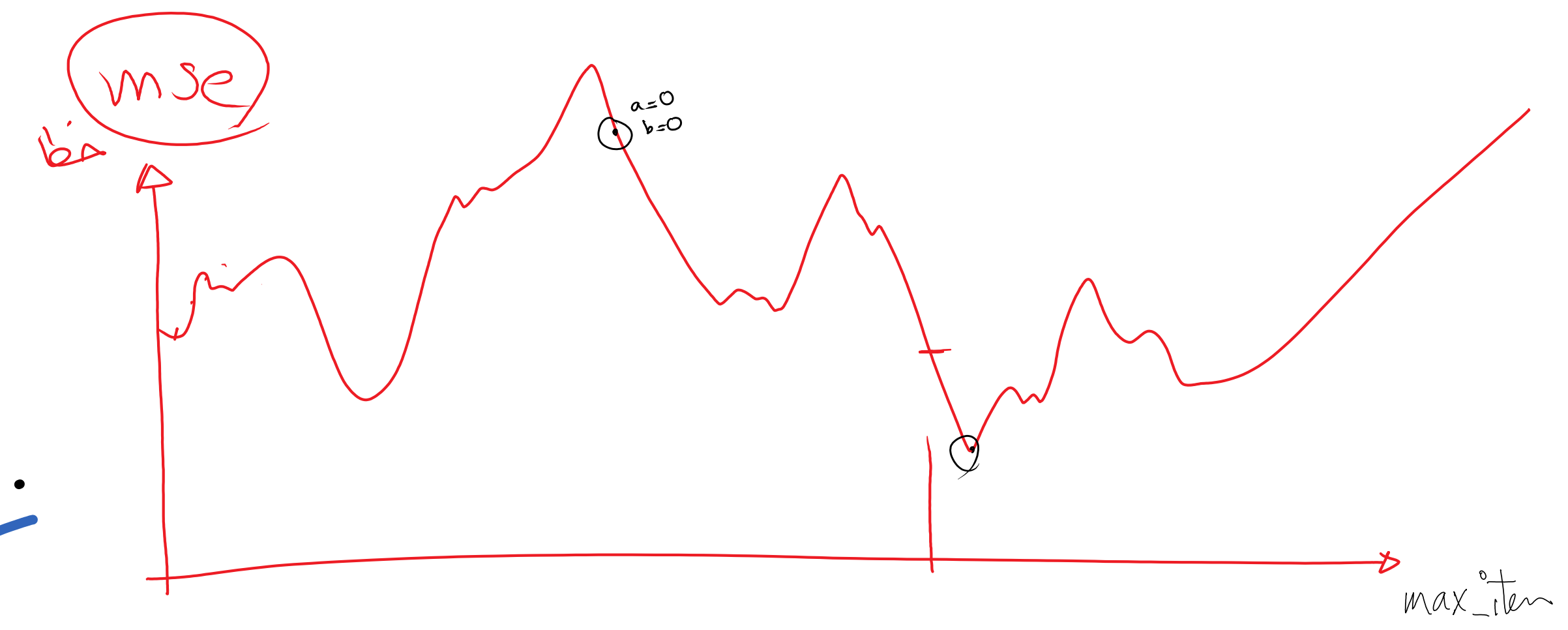
Param [1] b

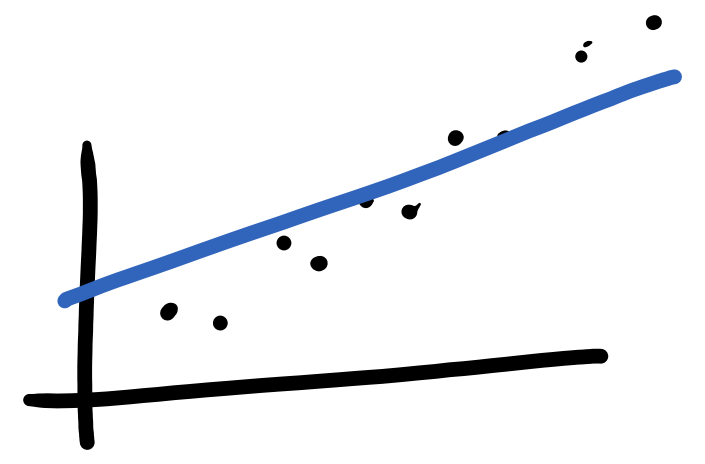


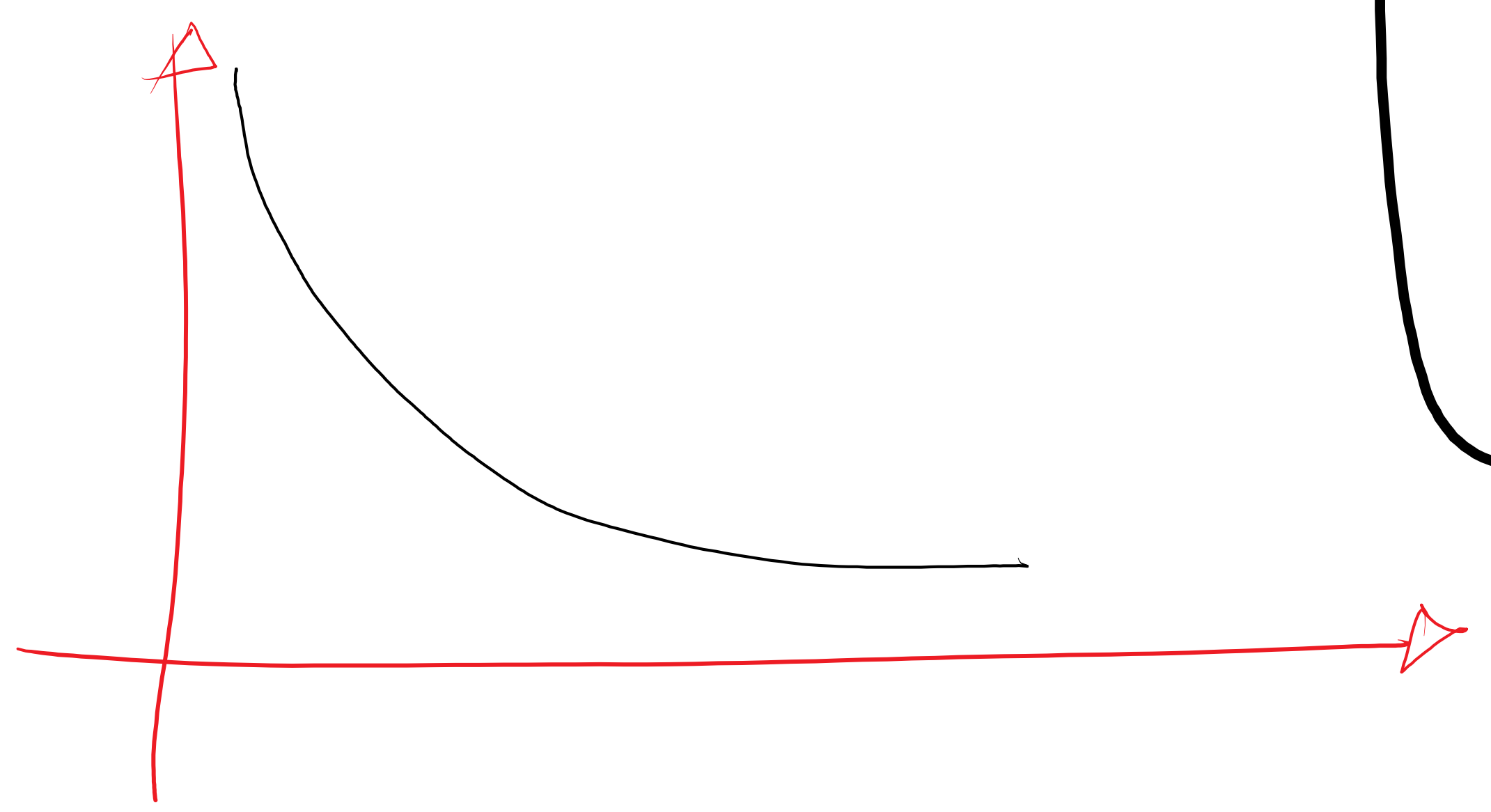
Input	α andβ calculated by GA
Output	Optimized coefficients α, andβ, and predicted data
1	Initialization of ε, θ, number of loops and partial derivative matrix
2	✓ Calculate the loss of the initial coefficients
3	do
4	{Calculate the newloss, i.e. the loss of the current coefficients
5	if(|newloss − loss| > d)
6	$\theta=\theta-\varepsilon\nabla J(\theta)$
7	Replace loss with newloss
8	}until (iterations > number of loops)
9	Get optimized coefficients

$\theta = \theta - \varepsilon \nabla j(\theta)$

(تابع هزینه)







$$\theta = \theta - \eta \boxed{\nabla(\theta)}$$